Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State the no-air-resistance assumption.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “free fall” and “initial” to explain your model or result. Sentence starters are optional.

free fall: motion with gravity as the only force

initial: at the beginning of the interval; subscript i

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A 4 kg cart starts from rest under 24 N net force right for 3 s.
Find acceleration and final velocity.
State one assumption.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

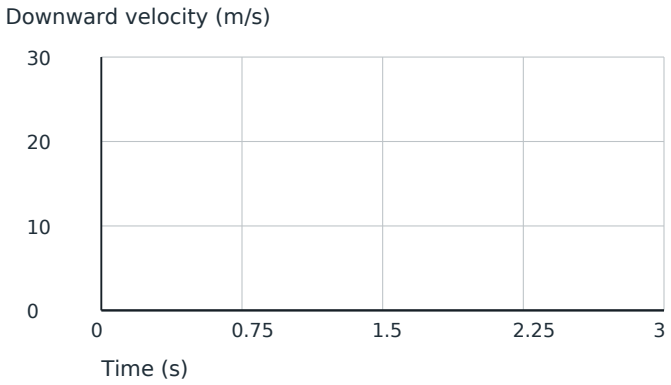
02 Worked example: explain the model

Reference: $v_f = v_i + g\,\Delta t$.
Downward positive; g = 10 m/s²; gravity only.

Worked reasoning: A ball dropped from rest has downward speeds 0, 10, 20 m/s at 0, 1, 2 s in ideal free fall with g=10 m/s². Equal times give equal velocity changes.

Your turn: Sketch its velocity-time graph with downward positive.
Explain why the positions are not equally spaced.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .



Marzano check - circle: ① ② ③

03 Read and interpret the model

Two balls of different mass are released together in a vacuum from the same height.
Compare acceleration and arrival time.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A ball is released from rest in a drop simulation. Ignore air resistance and use g = 10 m/s². Complete the table, then describe how the velocity changes each second.

Sentence starter: I notice … . This supports … because … .

Time (s)	Downward velocity (m/s)
0	0
1	____
2	____
3	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the claim.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A fixed-mass cart has acceleration 4 m/s² under 8 N net force.
Find acceleration when net force becomes 12 N.
Explain.
Use the table to keep the controlled and changed quantities distinct.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

07 Exit check

A dropped ball falls from rest for 4 s in an air-free simulation.
Use g = 10 m/s².
Find its final velocity and add a correctly directed arrow to the model.

Sentence starter: I found … [unit]. This value means … in this situation.

Dropped ball

State your positive direction.

Velocity:

Acceleration / force:

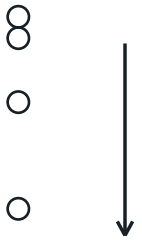
Marzano check - circle: ① ② ③

08 STRETCH: test the model

Draw three equal-time snapshots of a dropped ball. Then draw a heavier ball released with the same conditions in a vacuum.
Compare the patterns.

Sentence starter: Both … . They differ in … because … .

Equal time intervals



Marzano check - circle: ① ② ③